

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 12

PRICE AND OUTPUT DETERMINATION UNDER MONOPOLY

MEANING OF PURE MONOPOLY:

Monopoly is a market situation in which:

- (i) A single firm is having full control over the production and supply of a commodity / service. Thus there is no difference between firm and industry.
- (ii) There is no close substitute of the product in the whole market.
- (iii) Cross elasticity of demand for the product is zero.
- (iv) A monopolist firm has no competition with any other firm moreover, new entry in the business is not possible.
- (v) Because the total quantity of commodity depends upon the production of one firm only thus the Monopolist can affect the price by his own policy while increasing or decreasing the output and market supply.
- (vi) When the price is lowered down either due to increased production or for the purposes of enhancing sales, AR and MR curves start falling. Thus these curves are negatively sloped. It must be remembered that for a Monopoly firm, MR is always less than Price (AR) at all levels of output. Due to this reason MR curve always remains below the AR curve.

Thus we can say that: "A pure Monopoly is said to exist if there is one and only one seller in a well defined market".

According to Prof. Triffins, "Pure Monopoly is that where the cross elasticity of demand for monopoly product is zero".

Although in practical life, Monopoly is very rare yet sometime, the state through legislation, may allocate such an authority to any institution / department. In Pakistan WAPDA for electricity supply and Pakistan Railway on railway services have been given monopoly powers.

CAUSES OF MONOPOLY:

Monopoly comes into existence due to many reasons:

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1. NATURAL RESOURCES:

Some countries have specific climatic conditions, characteristics of the soil, ownership of raw material and mineral resources for the production of particular goods. Due to specialization in the production of related good a country has full control and monopoly over it's supply. e.g. jute of sub-continent, coffee of Brazil etc.

2. SOCIAL WELFARE:

In many economic activities competition is regarded wasteful. For the welfare and benefit of the members of the society such a competition may be avoided while creating Monopoly e.g. WAPDA and Railways department in Pakistan.

3. PATENT AND COPY RIGHTS:

An inventor, innovating firm, Author of a book etc. may be granted patent and copy rights by legal authority. NO other firm or person can utilize it.

4. TECHNICAL KNOW-HOW:

Some times a firm has special know-how or exclusive knowledge of production techniques about production of a commodity. Others have no access to such techniques. Hence the firm, commands Monopoly e.g. F-16 of U.S.A.

5. UNITY AMONG ENTREPRENEURS:

Monopolies are also created when some entrepreneurs agree upon some open or secret formula regarding the sale, Management or price of the commodity. Such agreements may be seen in the form of pool, cartel, trust etc.

6. HEAVY AND COSTLY PROJECTS:

There are some production fields where initial cost of the project is very heavy and even out of reach of many firms. Thus the firm who dares to enter the field may enjoy monopoly power e.g. Natural gas field.

BILATERAL MONOPOLY

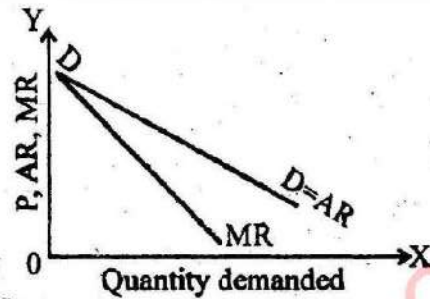
In a bilateral monopoly market there is only one producer / supplier (Monopolist) while on the other end there is only one buyer (Monopsonist) of the commodity or service. For instance, if only one producer is supplying copper, while on the other side there is only one manufacturer demanding raw copper, it will be regarded bilateral monopoly.

DEMAND CURVE UNDER MONOPOLY

As a Monopolist firm represents the whole industry. Hence AR the (demand curve) represents the both.

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The demand curve (AR) under monopoly is negatively sloped. The slope of demand curve $\frac{dq}{dp} = -b$, as shown in the following diagram.



AR = DD is a monopolist firms demand curve which moves downwardly from left to right. It is just like an ordinary market demand curve.

SHORT-RUN EQUILIBRIUM OF FIRM UNDER MONOPOLY

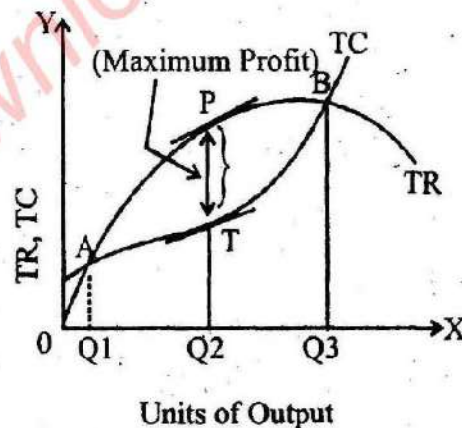
A Monopolist firm is also a profit oriented unit. It tries to produce the quantity of the product at which the profit is maximum or loss in minimum.

Short-run equilibrium of the firm may be discussed with the help of two approaches.

1. TOTAL REVENUE (TR) AND TOTAL COST (TC) APPROACH:

Under this approach firms equilibrium will be at the output level where difference between TR and TC is maximum. As $\pi = TR - TC$.

Following diagram represents the above mentioned method.



EXPLANATION:

TR and TC are short-run total revenue and total cost curves. $TR = TC$ at point A where output level is OQ_1 . At point B again $TR = TC$ where output is OQ_3 . It means that at these two points (A & B) firm is earning normal profit.

As below Q_1 and beyond Q_3 $TR < TC$ thus firm will have to face losses if production

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is less than OQ_1 or more than OQ_3 . PT shows the maximum difference between TR and TC. Under this situation output level is OQ_2 which indicates maximum profit and equilibrium level of the firm. Point P being the tangent point of TR, and T, tangent of TC. Thus at these two points slope of TR = Slope of TC and profit is maximum at output level OQ_2 .

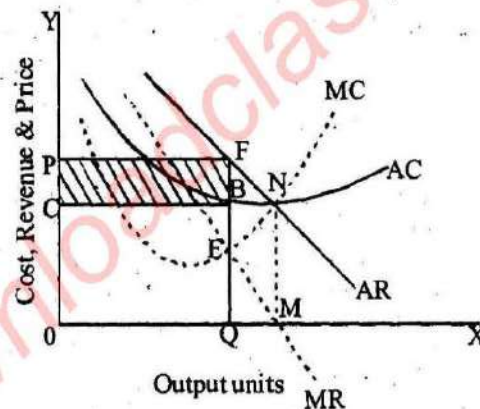
2. MR = MC APPROACH:

According to this approach the equilibrium level of a monopolist firm will be at the level of output where:

- (i) $MR = MC$
- (ii) MC intersects MR curve from below (Same conditions as under perfect competition)

The general assumption regarding a Monopolist firm is that it usually earns abnormal profit. It is due to no competition, no close substitute, no entry and full control over the supply of a commodity.

Following diagram shows the equilibrium level of a Monopolist firm earning "Abnormal profit"



In the diagram MC intersects MR curve at point "E" which is the equilibrium point. To know the output level we move vertically downward towards x-axis and reach at point Q. Thus OQ is the level of output while Monopolist firm is in equilibrium position.

To find out Average Cost (AC) and Average Revenue (AR) we move upward from point "E" and find point "B" on AC curve and point "F" on AR curve.

Thus at output OQ, $AC = BQ$ and $AR = FQ$

As $AR = FQ > AC = BQ$, Hence firm is earning abnormal profit.

$$\pi = TR - TC, TR = AR(OQ) = FQ(OQ) = OQFP$$

$$TC = BQ(OQ) = OQBC$$

$$\pi = OQFP - OQBC = PCBF \text{ (Shaded area)}$$

As $TR > TC$ thus this is the abnormal profit of the firm.

It must be noted that a Monopolist firm determines the price (P) of its product where

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$E > 1$ (Elasticity of demand at (AR) is greater than unity)

DOES A MONOPOLIST FIRM ALWAYS EARN ABNORMAL PROFIT?

A general concept regarding a Monopolist firm is that due to absence of close substitute and no competition with any firm, it charges higher price and always earns abnormal profit.

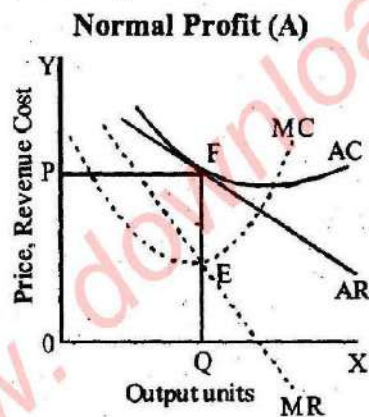
But this is not true. Even in short period due to certain threats or considerations a Monopolist firm will have to lower down the price of its product, although temporarily resulting normal profit position or short while losses.

The circumstances under which a Monopolist firm decreases the price of its product may be as under:

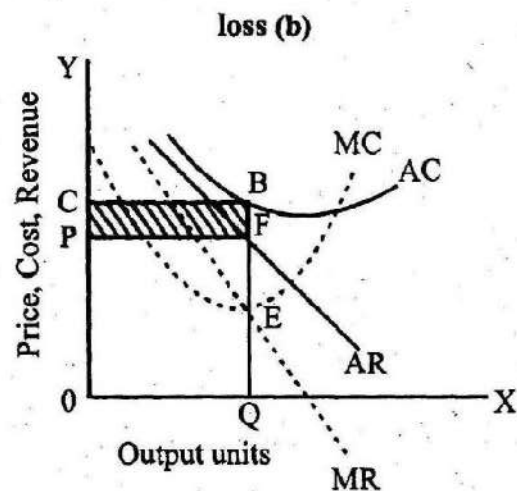
1. To avoid the danger of state interference or any action against the high price being charged by it.
2. To avoid the danger of substitute of its product.
3. To avoid boycott of the good in view of expecting unity among consumers.
4. Under "dumping policy" which may be due to over-production and more supply than demand in internal market. In order to keep the internal prices stable, Extra quantity is sold in foreign markets at even lower prices which is known as "Dumping".

In view of above mentioned possibilities, a Monopolist firm may lower down the price of its commodity upto the extent where $AR = AC$ (normal profit earning position) or $AR < AC$ (losses position). It may be time-being and temporary until the required aim of the firm is fulfilled.

Both the possibilities are explained graphically as under:



Above diagram shows the normal profit of the Monopolist firm at equilibrium point "E". It decreases the price of the output up to the extent where $AR = AC$.



Above diagram shows that Monopolist firm is under such a pressure that at the equilibrium level "E". The price of its product is less than AC. It means that at output level OQ firm's

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As $\pi = TR - TC$

and $TR = AR (Q) = FQ (OQ) = OQFP$

while $TC = AC (Q) FQ (OQ) = OQFP$

Here $TR = TC = OQFP$

Profit is normal as it is included in total cost (TC)

$AC > AR$ which shows

$TC > TR$ as $\pi = TR - TC$

$TR = FQ (OQ) = OQFP$

$TC = BQ (OQ) = OQBC$

$\pi = OQBC - OQFP = PFBC$ is the loss being faced by the firm.

It may be temporary but at present situation, it is a reality.

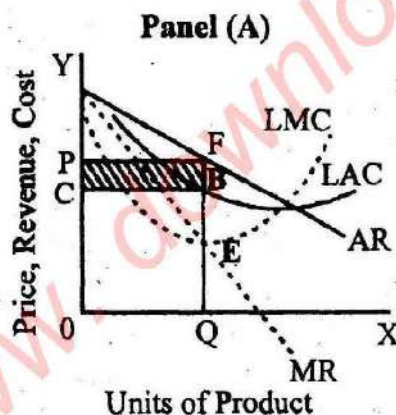
LONG-RUN EQUILIBRIUM OF THE FIRM UNDER MONOPOLY

Under Monopoly there is no possibility of new entry even in the long-run. Monopolist firm can extend or contract the size and production capacity of the plant in view of changes in market demand for its product.

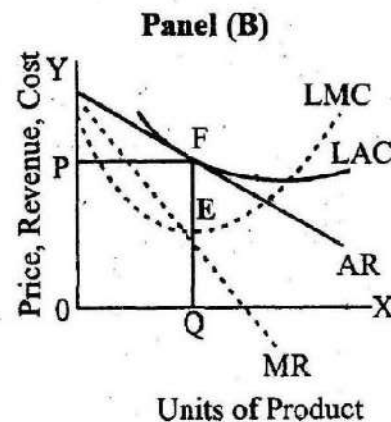
The size of profit depends upon the relationship between Price (P) and Long-run Average Cost (LAC) of the firm.

$LAC < P$ firm earns abnormal profit even in the long-run

In the following diagram panel (A) shows abnormal profit while normal profit has been shown in panel (B).



Panel (A) shows abnormal profit of a Monopolist firm. As at equilibrium point "E" output = OQ, $LAC = BQ$, $AR = FQ$
 As $LAC = BQ < AR = FQ$
 Thus $TC = OQ (BQ) = OQBC$
 $TR = OQ (FQ) = OQFP$



Panel (B) exhibits normal profit of a Monopolist firm. At equilibrium point "E" output = OQ
 $LAC = AR = FQ$
 $TR = FQ (OQ) = OQFP$
 $TC = FQ (OQ) = OQFP$

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$$\pi = TR - TC$$

$$\pi = OQFP - OQBC = \text{PCBF}$$

It is abnormal profit because

$TR > TC$ (Shaded area)

$$\pi = TR - TC \text{ as } TR = TC = OQFP$$

Thus firm in the long-run is earning normal profit.

WELFARE EFFECTS OF MONOPOLY

1. INTERNAL AND EXTERNAL ECONOMIES:

Large scale production encourages specialization, which provides internal and external economies. It results low cost and high profits.

2. PRICE STABILITY:

As a Monopolist firm have full control over the production and supply of a good. Thus supply can easily be adjusted according to market demand. The equilibrium between demand and supply forces keeps the prices stable.

3. INCREASE IN GOVT. REVENUE:

A Monopolist usually produces large quantity which pushes up Govt. Revenue due to sales tax, custom duty, income tax etc.

4. RESEARCH AND INNOVATION:

Due to heavy financial resources and capital availability, a monopolist can easily carry out research, discovery and innovation activities. New and cheaper production techniques can be adopted which may have healthy effect upon production costs and price level.

5. ECONOMY IN ADVERTISING:

In the absence of competition and close substitute there is no need of any type of propaganda and publicity. A reasonable amount is saved in this respect.

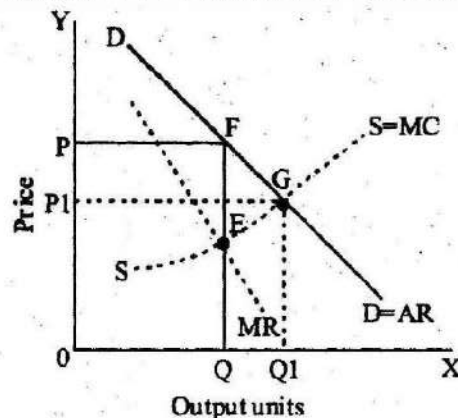
IN-EFFICIENCIES OR EVILS OF MONOPOLY

Following points indicate the weaknesses of a Monopolist firm.

1. IN-EFFICIENT USE OF RESOURCES:

The Monopolist does not use resources at their full capacity. She works at equilibrium level where $MC = MR$. Here price $P > MC$, output is less than the potential efficiency and profit becomes supernormal as shown in the following diagram at point 'E', where $Q = OQ$, $P = OP$ on the other side, maximum welfare of consumers, requires that quantity of output should be determined at market price in view of demand and supply forces as under perfect competition. In the following diagram $AR = DD$ while $ES = SS$ (Marginal cost curves above part of point E represents supply curve)

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(i) The Monopolist Approach:

MR = MC at point "E" where
 $Q = OQ$, $P = OP$ (Less output, high price)

(ii) Society Welfare Approach:

DD = SS at point "G" where
 $Q = OQ_1$, $P = OP_1$
 (More output, low price)

2. NO CHOICE FOR CONSUMERS:

A Monopolist firm produces the commodity of its own choice. Consumers' taste and liking is ignored.

3. UN-EQUAL DISTRIBUTION OF INCOME AND WEALTH:

The Monopolist usually charges abnormal profits. It causes greater inequality in distribution of income and wealth. The gap between rich and poor expands. Thus there is exploitation of consumers.

4. DISCOURAGEMENT OF NEW ENTRY:

The barriers to the entry of new firms discourage initiative, efficiency and modern know how and skill. It retards economic development.

5. NO INTEREST IN INNOVATIONS:

In the absence of rival firms, the monopolist feels very secure. Therefore, she does not care for new discoveries, innovations and technological progress. It is considered a hindrance to develop new methods and techniques of production.

6. ANTI-SOCIAL POLICIES:

The Monopolist produces the commodity giving high profits. It may be in the use of even a smaller group of people. Here the larger interest of the masses is ignored. A large number of consumers are deprived of basic needs and necessities. While only will of profit dominates and dictates.

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OBJECTIVE QUESTIONS

Select and tick (✓) the correct Answer.

1. Which firm controls the whole supply of a commodity:
(a) Monopolist (b) Duopolist (c) Bi-lateral (d) Monopsonist
2. Which among following represent Monopolies:
(a) Pakistan Railway (b) Commercial Banks
(c) Pakistan State Oil (d) WAPDA
3. Under Monopoly slope of MR curve—slope of AR curve:
(a) Less (b) Equal
(c) Greater (d) Same curve of the both
4. Under Monopoly in equilibrium elasticity of demand becomes:
(a) $E = 1$ (b) $E > 1$ (c) $E < 1$ (d) $E = 0$
5. Cross elasticity of product under Monopoly is:
(a) Zero (b) Infinite (c) $EC > 1$ (d) $EC < 1$
6. If there is only one producer (seller) and one buyer, the situation will be known as:
(a) Duopoly (b) Bi-lateral Monopoly
(c) Competitive firm (d) Monopsony
7. Under Monopoly when TR is maximum, MR is:
(a) Positive (b) Zero (c) Negative (d) $MR = 1$
8. When a Monopolist Firm charges higher price in local markets and lower price in external markets, it is known as:
(a) Dumping (b) Price discrimination
(c) Cartel (d) Pool
9. Which factor can be ignored by the Monopolist:
(a) Own Profit (b) Danger of Substitute
(c) Consumers Choice (d) Govt. Interference
10. Agreement of division of market in different parts under "entrepreneurs unity" is called:
(a) Trust (b) Pool (c) Cartel (d) Price Leadership

DESCRIPTIVE PART – I

SHORT NOTES

1. What is meant by Pure Monopoly?

A pure monopoly is said to exist when there is one and only one seller/producer in a well-defined market of a product or service. There is no close substitute of the product. A Monopolist can influence the price of the good/ service

2. Point out three circumstances under which monopoly situation may arise?

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- 1) Natural Resources
- 2) Technical Knowledge
- 3) Social welfare.

3. What is Bi-lateral Monopoly?

In bilateral monopoly market there is only one producer / supplier while on the other end there is only one buyer of the commodity.

4. The monopolist firm always earns abnormal profit Do you agree with statement. Give three arguments in favor your answer.

Generally monopolist firm earns abnormal profit but in certain conditions the firm reduces its price and earns normal profit i.g. (1) To avoid danger of state interference (2) To avoid the danger of substitute. (3) to avoid boycott by consumers.

5. What is the long run equilibrium position of a Monopolist firm?

Monopolist firm earn abnormal profit generally but in certain conditions it may earn normal profit in the long run.

6. Point out three Welfare effects of Monopoly?

- (i) Internal and External Economies
- (ii) Price Stability
- (iii) Increase in govt. revenue.

7. Underline four "Evils" of Monopoly?

- (i) In- Efficient use of resources
- (ii) No choice for consumers
- (iii) Un equal distribution of income and wealth
- (iv) Discouragement of new entry.

8. When and why a monopolist firm lowers down the price of the commodity?

- (i) To avoid danger of state interference
- (ii) To avoid danger of substitute
- (iii) To avoid boycott of consumers
- (iv) Under dumping

9. What is meant by Monopsonist.

In bilateral monopoly market there is only one producer (monopolist) while , only one buyer (monopsonist) of the commodity. When there is only one buyer of the good/ service, that is known as monopsonist.

10. What is Dumping?

If there is over production and more supply than demand in internal market then in order to keep internal price stable, extra quantity is sold in foreign markets at lower price which is known as Dumping.

DESCRIPTIVE PART – II

1. How price is determined under Monopoly.
2. Explain "Short-run" equilibrium position of a Monopolist firm.
3. Draw "Demand curve" under Monopoly.
4. Explain "Long-run" equilibrium of the Monopolist firm.

